

Mathematics A

Unit 2 • Chapter OL-T51 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

8 classified items • 38 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T51 • Expertise Q16+ • 8 items • 38 marks

Chapter	Items	Marks	Starts
01. OL-T51 Vectors	8	38	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Vectors

Unit 2 • Expertise • 38 marks • 8 item(s)

June 2025 | 4MA1-1H Q23 | 5 marks

Find an intersection vector from two paths

June 2024 | 4MA1-2H Q24 | 6 marks

Find an intersection vector from two paths

November 2023 | 4MA1-2H Q22 | 5 marks

Recover a division ratio using vector parameters

June 2023 | 4MA1-1H Q19 | 5 marks

Recover a division ratio using vector parameters

November 2024 | 4MA1-1H Q24 | 6 marks

Find an intersection vector from two paths

November 2024 | 4MA1-2H Q17 | 3 marks

Find magnitude of a vector

May-Nov 2020 | 2H Q19 | 3 marks

Use vectors in a parallelogram, trapezium or polygon path

January 2023 | 1H Q24 | 5 marks

Find a section or midpoint vector

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

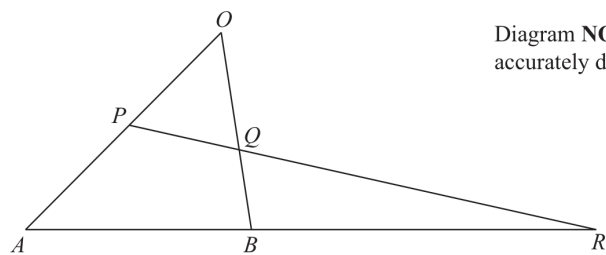


Diagram **NOT**
accurately drawn

OAB is a triangle.
 P is the midpoint of OA
 Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

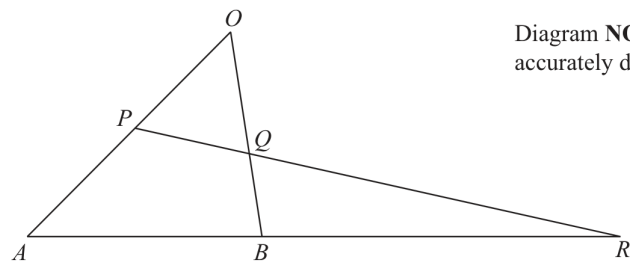
(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(1)

WORKING SPACE

4MA1-1H Q23

23

Diagram **NOT**
accurately drawn OAB is a triangle. P is the midpoint of OA Q is a point on OB ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

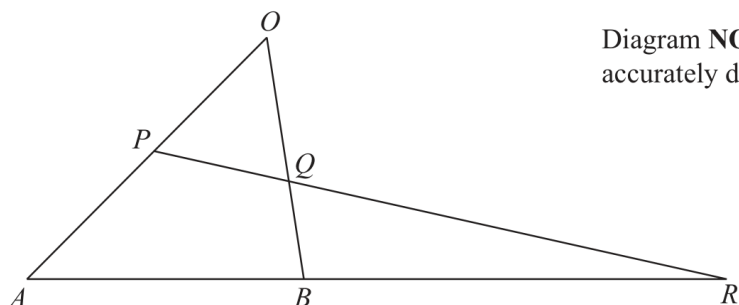
(b) Use a vector method to find the value of n

$$n = \dots\dots\dots (4)$$

WORKING SPACE

4MA1-1H Q23

23

Diagram **NOT**
accurately drawn OAB is a triangle. P is the midpoint of OA Q is a point on OB ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \dots\dots\dots (4)$$

Vector method on two straight lines

5 marks

Find $\vec{AB}$

$$\vec{AB} = \vec{OB} - \vec{OA} = 8\mathbf{b} - 12\mathbf{a}.$$

Find $\vec{PR}$

$$\vec{OR} = \vec{OA} + 3\vec{AB} = -24\mathbf{a} + 24\mathbf{b}, \quad \vec{OP} = 6\mathbf{a}, \quad \vec{PR} = -30\mathbf{a} + 24\mathbf{b}.$$

Use collinearity

$$\vec{OQ} = \vec{OP} + \lambda\vec{PR} = 6\mathbf{a} + \lambda(-30\mathbf{a} + 24\mathbf{b}). \quad 6 - 30\lambda = 0 \Rightarrow \lambda = \frac{1}{5}, \quad n = 24\lambda = \frac{24}{5}.$$

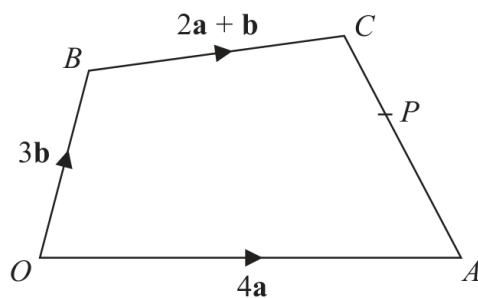
Final answer

$$n = \frac{24}{5}$$

4MA1-2H Q24

6 marks

24

Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

WORKING SPACE

4MA1-2H Q24

6 marks

24

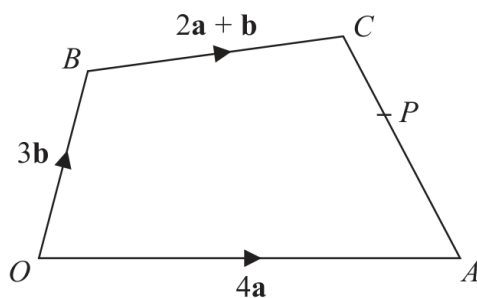


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
 Give your answer in its simplest form.
 Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

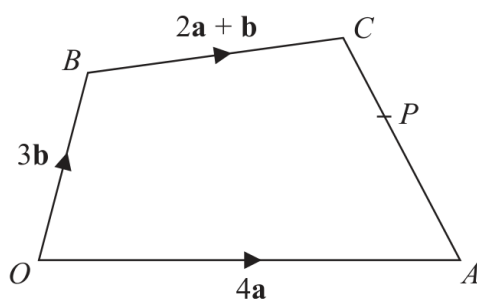
Continue your method**6 marks**

WORKING SPACE

4MA1-2H Q24

6 marks

24

Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\overrightarrow{OA} = 4\mathbf{a} \quad \overrightarrow{OB} = 3\mathbf{b} \quad \overrightarrow{BC} = 2\mathbf{a} + \mathbf{b}$$

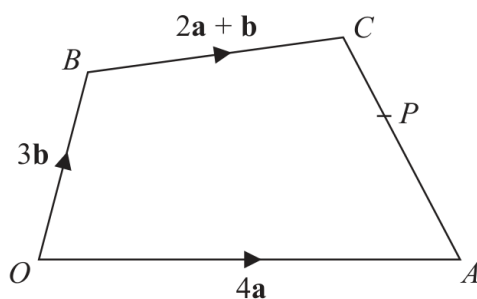
- (a) Find $\overrightarrow{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\overrightarrow{AC} = \dots\dots\dots (2)$$

4MA1-2H Q24

6 marks

24

Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

Vectors and collinearity

6 marks

Part (a)

$$\vec{AC} = 4b - 2a.$$

Point P

$$\vec{OP} = 4a + \frac{3}{5}(4b - 2a) = \frac{14}{5}a + \frac{12}{5}b.$$

Point Q

$$\vec{OQ} = \lambda \vec{OP} = 3b + \mu(2a + b) \Rightarrow \lambda = 3.$$

Final answer

$$\vec{OQ} = \frac{42}{5}a + \frac{36}{5}b$$

4MA1-2H Q22 (a)(i), (a)(ii), (b)

5 marks

22

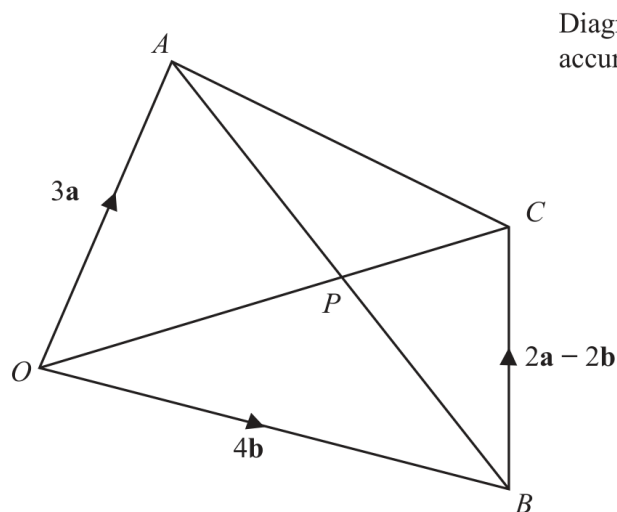


Diagram **NOT**
accurately drawn

$OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots (1)$$

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots (1)$$

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

$$\dots\dots\dots (3)$$

Continue your method

5 marks

WORKING SPACE

4MA1-2H Q22 (a)(i), (a)(ii), (b)

5 marks

22

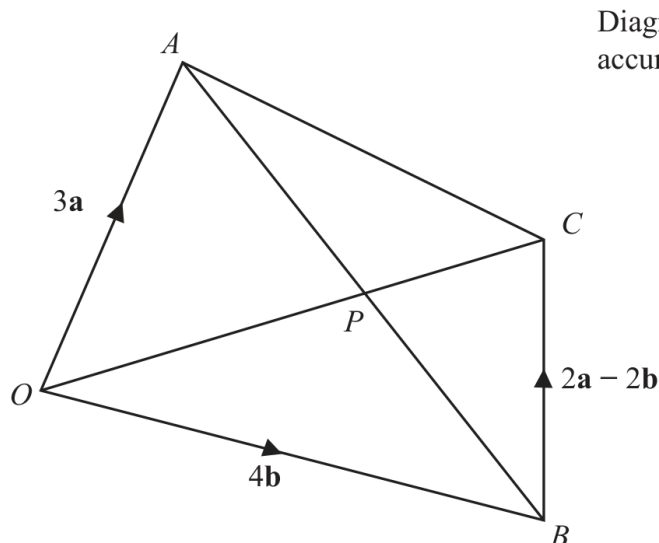


Diagram **NOT**
accurately drawn

$OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots$$

(1)

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots$$

(1)

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

$$\dots\dots\dots$$

(3)

Find OC from two vectors / Find AB from position vectors / Ratio where two vector descriptions meet **5 marks**

(a)(i) – Find OC from two vectors

Use the path O-B-C

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b.$$

Final answer

$$OC = 2a + 2b$$

(a)(ii) – Find AB from position vectors

Subtract position vectors

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a.$$

Final answer

$$AB = 4b - 3a$$

(b) – Ratio where two vector descriptions meet

Describe P on both lines

$$\overrightarrow{OP} = \lambda(2a + 2b) = 3a + \mu(4b - 3a).$$

Compare coefficients

$$2\lambda = 4\mu, \quad 2\lambda = 3 - 3\mu \Rightarrow \mu = \frac{3}{7}.$$

Convert the parameter to a ratio

$$AP : PB = \mu : (1 - \mu) = \frac{3}{7} : \frac{4}{7} = 3 : 4.$$

Final answer

$$AP:PB = 3:4$$

4MA1-1H Q19

5 marks

19

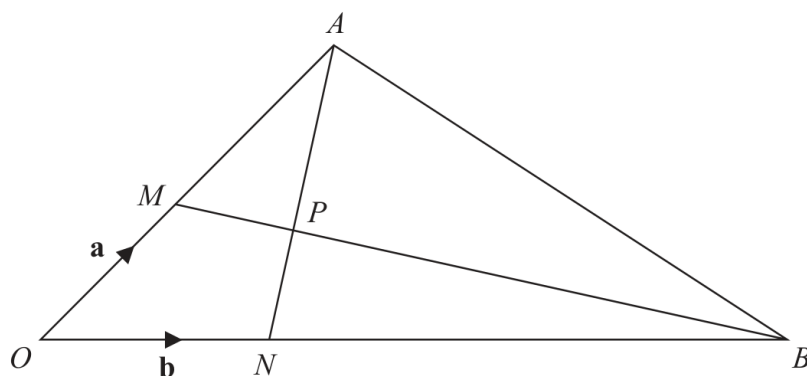


Diagram **NOT**
accurately drawn

OMA , ONB , MPB and NPA are straight lines.

M is the midpoint of OA

$ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vector $\vec{AN}$

.....
(1)

(b) Use a vector method to find the ratio $AP:PN$

$AP:PN =$
(4)

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q19

5 marks

19

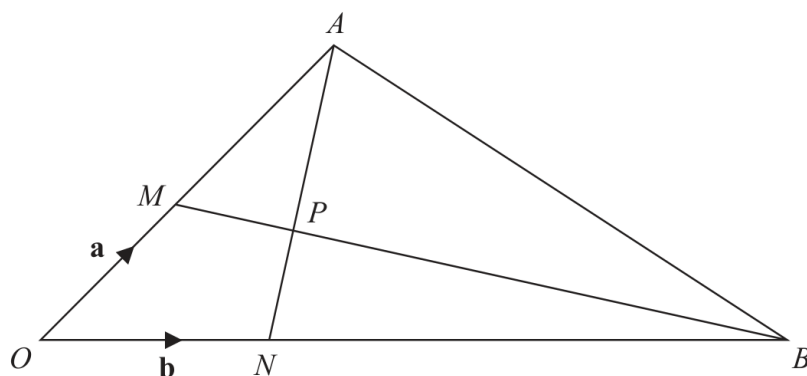


Diagram **NOT**
accurately drawn

OMA , ONB , MPB and NPA are straight lines.

M is the midpoint of OA

$ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vector $\vec{AN}$

.....
(1)

(b) Use a vector method to find the ratio $AP:PN$

$AP:PN =$
(4)

Intersection ratio by vectors

5 marks

Write the key vectors

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 6\mathbf{b}, \quad \overrightarrow{AN} = \mathbf{b} - 2\mathbf{a}.$$

Describe P along AN

$$\overrightarrow{OP} = 2\mathbf{a} + \lambda(\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + \lambda\mathbf{b}.$$

Describe P along MB

$$\overrightarrow{OP} = \mathbf{a} + \mu(6\mathbf{b} - \mathbf{a}) = (1 - \mu)\mathbf{a} + 6\mu\mathbf{b}.$$

Equate coefficients

$$\lambda = 6\mu, \quad 2 - 2\lambda = 1 - \mu \Rightarrow \lambda = \frac{6}{11}.$$

Final answer

$$\overrightarrow{AN} = \mathbf{b} - 2\mathbf{a}; \quad AP : PN = 6 : 5$$

4MA1-1H Q24 M01, M02, M03

6 marks

24 OAB is a triangle.

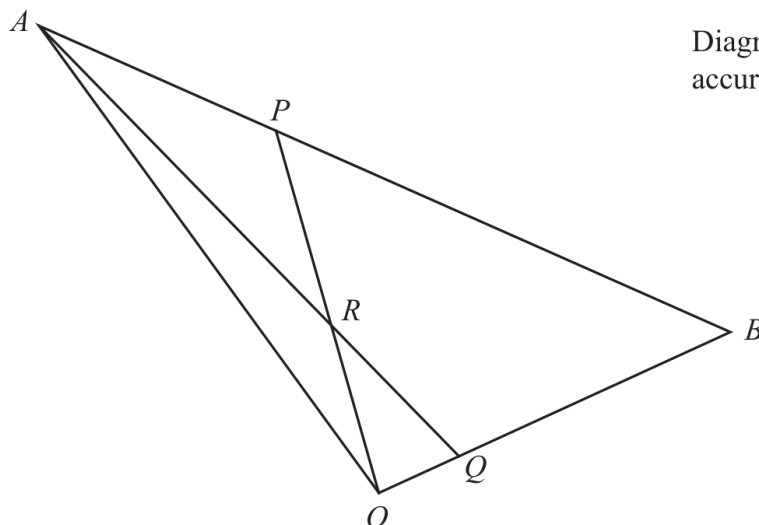


Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

4MA1-1H Q24 M01, M02, M03

6 marks

(iii) $\vec{OR}$
(4)

WORKING SPACE

4MA1-1H Q24 M01, M02, M03

6 marks

24 OAB is a triangle.

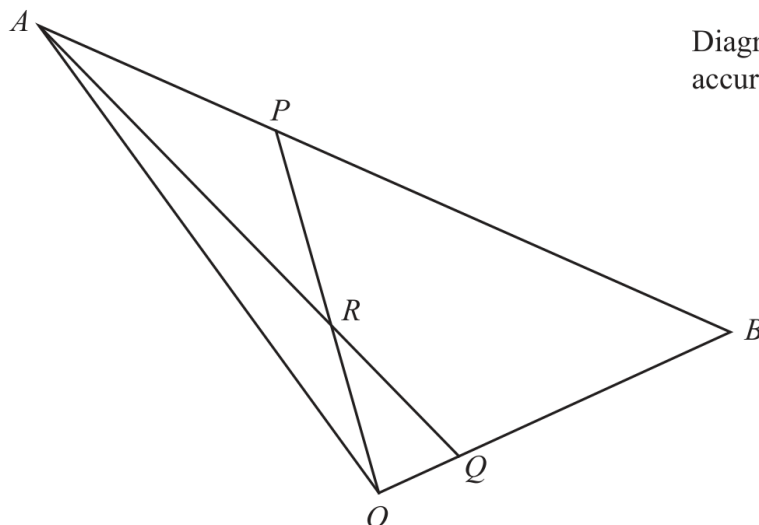


Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

4MA1-1H Q24 M01, M02, M03

6 marks

(iii) $\vec{OR}$
(4)

Find AQ / Find OP / Find OR from two straight-line routes

6 marks

M01 – Find AQ

Use a route through O

$$\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ} = -10\mathbf{a} + 2\mathbf{b}.$$

Final answer

$$-10\mathbf{a} + 2\mathbf{b}$$

M02 – Find OP

Use AP=AB/4

$$\overrightarrow{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a}) = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}.$$

Add OA

$$\overrightarrow{OP} = 10\mathbf{a} + \overrightarrow{AP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}.$$

Final answer

$$\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

M03 – Find OR from two straight-line routes

Write OR through P

$$\overrightarrow{OR} = \overrightarrow{OP} + \lambda\overrightarrow{PR}.$$

Write OR through Q

Use the line ARQ and the known AQ route to form a second expression.

Equate coefficients

The two routes give $\lambda=1/2$ and hence vector $\overrightarrow{OR}=(15)/(4)\mathbf{a}+5/4\mathbf{b}$.

Simplify

$$\overrightarrow{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}.$$

$$\frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

4MA1-2H Q17 M01

3 marks

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

.....

WORKING SPACE

Find the magnitude of HF

3 marks

M01 – Find the magnitude of HF**Find the vector**

$$\overrightarrow{HF} = \overrightarrow{HG} - \overrightarrow{FG} = (4, 14) - (-5, 2) = (9, 12).$$

Use Pythagoras

$$|HF| = \sqrt{9^2 + 12^2} = 15.$$

State the magnitude

$$|HF| = 15.$$

Final answer

15

19 OAB is a triangle.

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

The point C lies on OA such that $OC : CA = 1 : 2$

The point D lies on OB such that $OD : DB = 1 : 2$

Using a vector method, prove that $ABDC$ is a trapezium.

(Total for Question 19 is 3 marks)

Answer reference | May-Nov 2020 | Question 19

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

19	$\overrightarrow{AB} = -\mathbf{a} + \mathbf{b}$ or $\overrightarrow{BA} = \mathbf{a} - \mathbf{b}$			M1	Correct diagram (condone missing vector labels or arrows – with C on line segment OA and D on line segment OB) OR for finding $\overrightarrow{AB}$ or $\overrightarrow{BA}$ - may be seen as part of later working
	$\overrightarrow{CD} = \frac{1}{3}(-\mathbf{a} + \mathbf{b})$ or $\overrightarrow{DC} = \frac{1}{3}(\mathbf{a} - \mathbf{b})$ oe			M1	Method to find $\overrightarrow{CD}$ or $\overrightarrow{DC}$
		Correct vectors and conclusion including <u>parallel</u> and <u>trapezium</u>	3	A1	eg $\overrightarrow{AB}$ (AB) and $\overrightarrow{CD}$ (CD) are parallel therefore $ABDC$ is a trapezium
					Total 3 marks

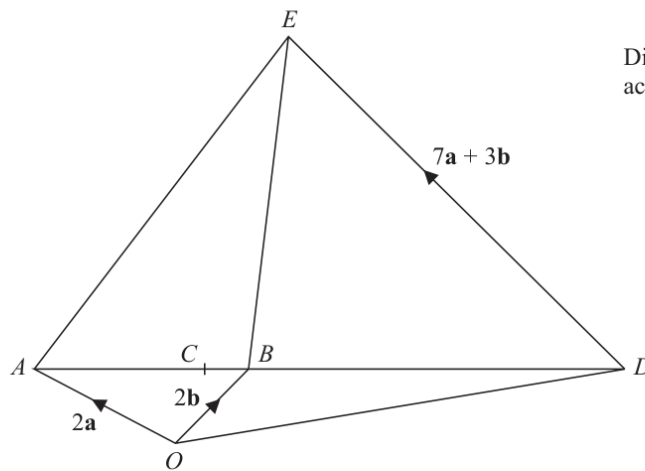
Worked solution

May-Nov 2020 | Question 19 | 3 marks

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

Official final mark-scheme pages are embedded immediately after this question.

Diagram **NOT**
accurately drawn



$$\overrightarrow{OA} = 2\mathbf{a} \quad \overrightarrow{OB} = 2\mathbf{b} \quad \overrightarrow{DE} = 7\mathbf{a} + 3\mathbf{b}$$

$$AB:BD = 1:2$$

The point C on AB is such that OCE is a straight line.

Use a vector method to find the ratio of $OC:CE$



(Total for Question 24 is 5 marks)

Answer reference | January 2023 | Question 24

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

24	e.g. $(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or $(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or $(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or $(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe		5	M1
	e.g. $(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe or $(\overrightarrow{OC}) = 2\mathbf{a} + \lambda(2\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b}$ oe or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b})$ or $(\overrightarrow{CE}) = (2\mathbf{b} - 2\mathbf{a}) - \lambda(2\mathbf{b} - 2\mathbf{a}) + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} = (1 + 2\lambda)\mathbf{a} + (9 - 2\lambda)\mathbf{b}$			M2 for 2 correct paths seen M1 for 1 correct path seen Any correct path for OC passing through A or B involving a variable
	e.g. $\frac{2 - 2\lambda}{2\lambda} = \frac{1 + 2\lambda}{9 - 2\lambda}$ oe or $\frac{2 - 2\lambda}{2\lambda} = \frac{3}{9}$ oe or $\frac{(1 + 2\lambda)}{(9 - 2\lambda)} = \frac{1}{3}$ oe or $\lambda = \frac{3}{4}$ or $(2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b} = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{3}{4}$ or $\mu = \frac{1}{6}$ or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b}) = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$			M1 for comparing coefficients of $\mathbf{a}$ and $\mathbf{b}$ for $(OC \text{ and } CE)$ or $(OC \text{ and } OE)$ or $(CE \text{ and } OE)$ OC is a multiple of OE Two different paths for OC
		1 : 5		A1 dep M2 oe e.g 2 : 10
	Working required			Total 5 marks

Answer reference | January 2023 | Question 24

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

24 ALT	e.g. $(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or $(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or $(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or $(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe		5	M1
	e.g. $(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe			M1
	e.g. $(\overrightarrow{AE}) = 2\mathbf{b} - 2\mathbf{a} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= \mathbf{a} + 9\mathbf{b})$ oe			M1
	$[\overrightarrow{AE} = \lambda \overrightarrow{AD} + \mu \overrightarrow{OE}]$ $\mathbf{a} + 9\mathbf{b} = \lambda(6\mathbf{b} - 6\mathbf{a}) + \mu(3\mathbf{a} + 9\mathbf{b})$ oe or $1 = -6\lambda + 3\mu$ oe and $9 = 6\lambda + 9\mu$ oe or or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$			M1
		1 : 5		A1 dep on M2 oe e.g 2 : 10
	Working required			Total 5 marks

Worked solution

January 2023 | Question 24 | 5 marks

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

Official final mark-scheme pages are embedded immediately after this question.